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| CONTACT INFORMATION | National Institute of Informatics<br>2-1-2 Hitotsubashi Chiyoda-ku Tokyo-to<br>101-8430 Japan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | E-mail: kphalakarn@nii.ac.jp<br>kphalakarn@gmail.com<br>Website: <a href="https://kphalakarn.github.io/">https://kphalakarn.github.io/</a> |
| RESEARCH INTERESTS  | <ul style="list-style-type: none"><li>• <i>Probabilistic model checking</i>: quantitative properties of Markov models and games</li><li>• <i>Lattice theory and fixed point computation</i>: value iteration, fixed point uniqueness</li><li>• <i>Algorithm design and analysis</i>: graph-based algorithms</li></ul>                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                            |
| EDUCATION           | <b>University of Waterloo</b> , Ontario, Canada<br>Ph.D., Electrical and Computer Engineering, 2019–2023<br><br><b>Chulalongkorn University</b> , Bangkok, Thailand<br>M.Eng., Computer Engineering, 2017–2019<br>B.Eng., Computer Engineering (First Class Honors), 2013–2017                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                            |
| HONORS AND AWARDS   | Best Paper Award, ICTAC 2024<br>University of Waterloo Faculty of Engineering Graduate Scholarship, 2023<br>Ripple Graduate Fellowship, 2019–2023<br>Chulalongkorn University Department of Computer Engineering Graduate Fellowship, 2017–2019<br>Chulalongkorn University Faculty of Engineering Gold Medal of Excellence, 2017<br>Outstanding Academic Performance Award, Engineering Institute of Thailand, 2016<br>First Solution Award, ACM-ICPC World Finals 2016                                                                                                                                                                                                         |                                                                                                                                            |
| ACADEMIC EXPERIENCE | <b>National Institute of Informatics</b> , Tokyo, Japan<br><i>Researcher</i> , Research Center for Mathematical Trust in Software and Systems (09.2023–present)<br><i>Research Intern</i> , ERATO Metamathematics for Systems Design Project (03.2019–08.2019)<br><br><b>RWTH Aachen University</b> , Aachen, Germany<br><i>Visiting Researcher</i> , Software Modeling and Verification Group (10.2025)<br><br><b>The University of Tokyo</b> , Tokyo, Japan<br><i>Research Intern</i> , Imai Laboratory, Department of Computer Science (06.2016–07.2016)                                                                                                                      |                                                                                                                                            |
| TEACHING EXPERIENCE | <b>University of Waterloo</b> , Ontario, Canada<br><i>Teaching Assistant</i> <ul style="list-style-type: none"><li>• ECE 606 Algorithm Design and Analysis: Fall 2020</li><li>• ECE 124 Digital Circuits and Systems: Spring 2020</li></ul><br><b>Chulalongkorn University</b> , Bangkok, Thailand<br><i>Teaching Assistant</i> <ul style="list-style-type: none"><li>• 2110201 Computer Engineering Mathematics (Linear Algebra): Winter 2019</li><li>• 2110202 Discrete Structures and Computability (Discrete Mathematics): Fall 2018</li><li>• 2110101 Computer Programming: Winter 2015, Fall 2016, Winter 2017, Spring 2017, Fall 2017, Winter 2018, Spring 2018</li></ul> |                                                                                                                                            |

**K. Phalakarn**, S. Pruekprasert, and I. Hasuo. “Strategy Templates for Almost-Sure and Positive Winning of Stochastic Parity Games towards Permissive and Resilient Control,” *Theoretical Computer Science (TCS)*, vol. 1057, no. 115535, 2025, pp. 1–15.

**K. Phalakarn**, Y.C. Tsai, and I. Hasuo. “Widest Path Games and Maximality Inheritance in Bounded Value Iteration for Stochastic Games,” *Proc. of the 23rd International Symposium on Automated Technology for Verification and Analysis (ATVA 2025)*, pp. 109–131.

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**K. Phalakarn**, V. Suppakitpaisarn, F. Rodríguez-Henríquez, and M. A. Hasan. “Vectorized and Parallel Computation of Large Smooth-Degree Isogenies using Precedence-Constrained Scheduling,” *IACR Trans. on Cryptographic Hardware and Embedded Systems (TCHES)*, vol. 2023, issue 3, pp. 246–269.

**K. Phalakarn**, V. Suppakitpaisarn, and M. A. Hasan. “Speeding-Up Parallel Computation of Large Smooth-Degree Isogeny Using Precedence-Constrained Scheduling,” *Proc. of the 27th Australasian Conference on Information Security and Privacy (ACISP 2022)*, pp. 309–331.

**K. Phalakarn**, V. Suppakitpaisarn, and M. A. Hasan. “Single-round Lattice-based Multisignatures,” *Proc. of the 8th International Workshop on Information and Communication Security (WICS 2021)*, pp. 365–371.

**K. Phalakarn**, T. Takisaka, T. Haas, and I. Hasuo. “Widest Paths and Global Propagation in Bounded Value Iteration for Stochastic Games,” *Proc. of the 32nd International Conference on Computer Aided Verification (CAV 2020)*, pp. 349–371.

**K. Phalakarn**, K. Phalakarn, and V. Suppakitpaisarn. “Optimal Representation for Right-to-Left Parallel Scalar and Multi-Scalar Point Multiplication,” *International Journal of Networking and Computing (IJNC)*, vol. 8, no. 2, 2018, pp. 166–185.

**K. Phalakarn**, and A. Surarerks. “A Matrix Decomposition Method for Odd-Type Gaussian Normal Basis Multiplication,” *Proc. of the 3rd International Conference on Computer and Communication Systems (ICCCS 2018)*, pp. 99–103.

**K. Phalakarn**, K. Phalakarn, and V. Suppakitpaisarn. “Optimal Representation for Right-to-Left Parallel Scalar Point Multiplication,” *Proc. of the 4th International Workshop on Information and Communication Security (WICS 2017)*, pp. 482–488.

**K. Phalakarn**, and A. Surarerks. “An Analysis of Computer Programs using  $\lambda$ -calculus,” *Proc. of the 7th International Workshop on Computer Science and Engineering (WCSE 2017)*, pp. 214–218.

K. Phalakarn, **K. Phalakarn**, and V. Suppakitpaisarn. “Parallelized Side-Channel Attack Resisted Scalar Multiplication Using  $q$ -Based Addition-Subtraction  $k$ -chains,” *Proc. of the 4th International Symposium on Computing and Networking (CANDAR 2016)*, pp. 140–146.

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| PATENTS                  | <b>K. Phalakarn</b> , T. Takisaka, T. Haas, and I. Hasuo. “System Optimal Control Device, System Optimal Control Method, and Program,” U.S. Patent 12 360 519, July 15, 2025.                                                                                                 |
| OTHER<br>PUBLICATIONS    | <b>K. Phalakarn</b> , K. Phalakarn, S. Prasitjutrakul, and S. Sinthupinyo. “Python 101,” Textbook for 2110101 Computer Programming course (in Thai), 2017.                                                                                                                    |
| PROFESSIONAL<br>SERVICES | <ul style="list-style-type: none"> <li>• Reviewer: FMCAD 2026, CONCUR 2026, ATVA 2023</li> <li>• Artifact Evaluation Committee: ATVA 2026, CAV 2026, FM 2026, VMCAI 2026</li> <li>• Session Chair: ATVA 2025</li> </ul>                                                       |
| SKILLS                   | <ul style="list-style-type: none"> <li>• Programming Languages: Python, C/C++, Java; some experiences with PRISM model checker, R, VHDL, Verilog, OpenMP API for parallel programming.</li> <li>• Languages: Thai (native), English (fluent), Japanese (beginner).</li> </ul> |